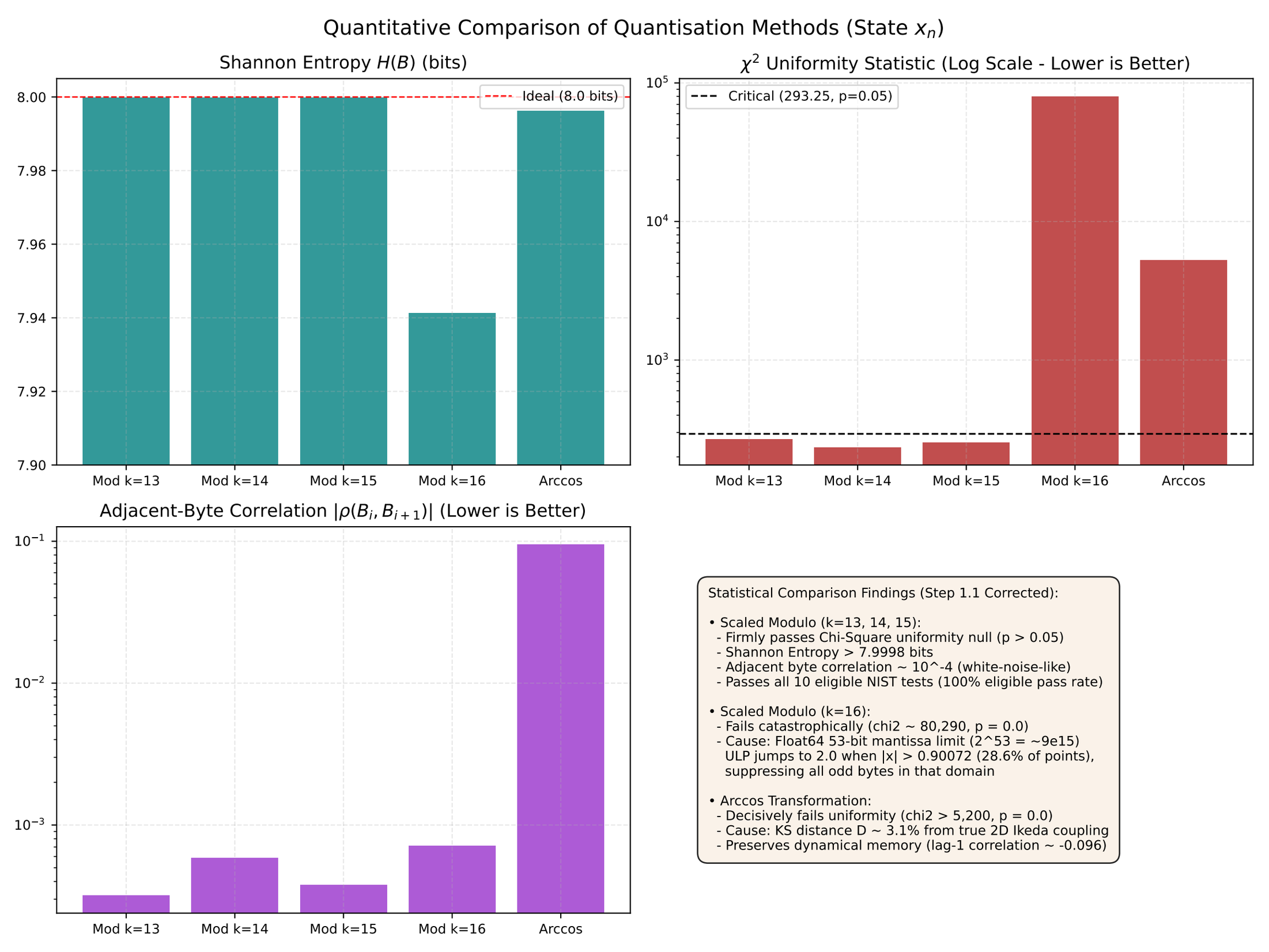
Quantitative Comparison of Quantisation Methods (State x_n)
Shannon Entropy $H(B)$ (bits)
Ideal (8.0 bits)
8.00
7.98
7.96
7.94
7.92
7.90
Mod k=13
Mod k=14
Mod k=15
Mod k=16
Arccos
χ^2 Uniformity Statistic (Log Scale - Lower is Better)
Critical (293.25, p=0.05)
10^5
10^4
10^3
Mod k=13
Mod k=14
Mod k=15
Mod k=16
Arccos
Adjacent-Byte Correlation $|\rho(B_i, B_{i+1})|$ (Lower is Better)
10^{-1}
10^{-2}
10^{-3}
Mod k=13
Mod k=14
Mod k=15
Mod k=16
Arccos
Statistical Comparison Findings (Step 1.1 Corrected):
• Scaled Modulo (k=13, 14, 15):
- Firmly passes Chi-Square uniformity null (p > 0.05)
- Shannon Entropy > 7.9998 bits
- Adjacent byte correlation ~ 10^-4 (white-noise-like)
- Passes all 10 eligible NIST tests (100% eligible pass rate)
• Scaled Modulo (k=16):
- Fails catastrophically (chi2 ~ 80,290, p = 0.0)
- Cause: Float64 53-bit mantissa limit (2^53 = ~9e15)
ULP jumps to 2.0 when |x| > 0.90072 (28.6% of points),
suppressing all odd bytes in that domain
• Arccos Transformation:
- Decisively fails uniformity (chi2 > 5,200, p = 0.0)
- Cause: KS distance D ~ 3.1% from true 2D Ikeda coupling
- Preserves dynamical memory (lag-1 correlation ~ -0.096)